

TANMOY ROY *PhD*

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SUMMARY

AI and Machine Learning professional with 19+ years of experience across industry and academia. PhD in Speech Emotion Recognition from University of Johannesburg. Currently Senior Data Scientist at DeliverHealth Solutions, leading AI-driven healthcare solutions including Medical Coding Co-pilot and Automatic Medical Note Generation. Prior experience at SIEMENS (8+ years) in telecom and smart-grid systems using C/C++, Java, and Oracle.

EXPERIENCE

DeliverHealth Solutions

REMOTE, INDIA

[JULY '24 – PRESENT]

Senior Data Scientist

Led research and development for AI-driven product and solution ideation, driving organization-wide AI adoption initiatives.

Collaborated with cross-functional stakeholders, including Product Owners and partners such as Google and Microsoft, to design and deploy scalable AI solutions addressing critical business challenges.

Key projects: Medical Coding Co-pilot (SLM finetuning), Automatic Medical Note Generation (Gemini Pro), RAG-based Medical Correction Engine, and Med-ASR finetuning (12% Medical WER).

Cloudcraftz Solutions

KOLKATA, INDIA

[NOV '22 – JUNE '24]

Senior Data Scientist

Primary responsibility was to lead and mentor a young team having diversified skills in Data Science, Quantitative Analytics and Software Engineering, and deliver complex solutions primarily for the Finance domain.

Key projects: Option Trading Strategy Backtest and Analytics, Explainable AI (SHAP/ALE), and Data Visualization Platform.

Veersa Technologies

KOLKATA, INDIA

[MAY '22 – OCT 22]

Data Scientist (Consulting)

Primary responsibility was ML model building with both tabular and image data. Other responsibilities involve data engineering, ML model productionizing, and ML team building. Executed projects in the healthcare domain.

Key projects: Insurance Claim Denial Prediction (XGBoost, 89% accuracy), Patient Data De-Duplication, Disease Detection from X-Ray (ResNet34, 84%), Clinical Trial Enrollment Prediction, and Handwritten Medical Form Reader (AWS Textract).

University of Johannesburg

JOHANNESBURG, SOUTH AFRICA

[JUL '16 – SEP '21]

Post Doctoral Research Fellow

[OCT '19 – SEP '21]

Full-time funded research position (GES fellowship): deep learning models for speech emotion recognition and two international research collaborations.

Doctoral Research Fellow

[JUL '16 – JUN '19]

Full-time funded doctoral position (GES scholarship): PhD completed in the stipulated three years on machine learning for speech emotion recognition.

SIEMENS

KOLKATA, INDIA

[Jan '07 – Sep '15]

Associate Consultant

[Jan '13 – Sep '15]

Telecom Service Provisioning, CRM integration, and GSM-R Billing systems. Technologies: Java Spring, C/C++, Oracle PL/SQL, Linux.

Sr. System Engineer

[Jan '11 – Dec '12]

Electronic Car billing (C/C++, Oracle) and Telecom Call Analysis GUI (Qt/C++).

System Engineer

[Jan '09 – Dec '10]

Telecom Billing and Interconnect Charging modules. Technologies: Java, ProC/C++, Oracle.

Associate System Engineer

[Jan '07 – Dec '08]

IVR/USSD system support and automated monitoring. Technologies: Java, C++, Linux, shell/awk scripts.

TECHNICAL EXPERTISE

- **Programming Languages:** Python, C/C++, R, Matlab, Java, JavaScript, SQL, Scala, Julia.
 - **Libraries:** Few prominent python libraries are: ollama, langchain, langgraph, unsloth, pytorch, pandas, matplotlib, scipy, opencv, pyspark.
 - **Technologies/Platforms:** Cloud (AWS, GCP), Linux, Docker container, MLOps (KubeFlow), REST, SOAP, XML.
 - **Databases:** RDBMS (Oracle, PostgreSQL, SQL Server, MySQL), NoSQL (MongoDB, DynamoDB, Cassandra), Cloud DB (Amazon RDS).
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EDUCATION

Doctor of Philosophy – University of Johannesburg, 2019. Applied ML & Speech Signal Processing.

Master of Science – Lovely Professional University, 2013. Information Technology.

PG Diploma in Computer Applications – DOEACC Society, 2004.

Bachelor of Science – University of Calcutta, 2002. Economics(Hons), CS, Mathematics.

CERTIFICATES

1. **Deep Learning with PyTorch : Neural Style Transfer**
Coursera
 2. **Neural Network from Scratch in TensorFlow**
Coursera
 3. **Billing and Revenue Management**
Oracle University
 4. **Diploma in Computer Networking**
CMC Limited
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REFERENCES

1. **Lance Mansfield** ([LinkedIn](#))
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2. **Nimish Sanghi** ([LinkedIn](#))

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